

Marc Berthelette

Senior Platform Engineer · Backend Systems · Reliability Engineering

Smiths Falls, ON, Canada · Remote · marc.berthelette@gmail.com

linkedin.com/in/marc-berthelette-3aba6917 · Bilingual: French (native) · English (fluent) · Eligible to work in Canada · Available immediately

Professional Summary

Senior platform and software engineer with 18 years of experience spanning production application development, backend services, infrastructure automation, and reliability engineering. At Desjardins, designed and built a Python/FastAPI service that continuously validated approximately 250 enterprise repositories and became the regression safety net for a platform upgrade. At Énergir, independently reverse-engineered nine application ecosystems and standardized delivery for 12 teams and 70+ applications. Earlier, wrote production C++ and Java software for approximately seven years before moving into platform engineering. Brings Python, FastAPI, Terraform, Docker, Kubernetes, AWS exposure, observability, and a record of taking ambiguous technical problems from investigation through production delivery.

Technical Skills

Backend and Software Engineering	Python, FastAPI, Java, C++, SQL, NoSQL, REST services, application integration, production debugging
Platform and Reliability Engineering	Service health automation, incident response, root-cause analysis, automated remediation, CI/CD, GitOps, technical design
Cloud and Infrastructure	Microsoft Azure, AWS (EC2, S3, Route 53, VPC), Terraform, Ansible, Puppet, Linux
Containers and Delivery	Docker, Kubernetes, ArgoCD, GitHub Actions, Azure DevOps, Jenkins, JFrog Artifactory
Observability	Dynatrace, Splunk, ELK Stack, PRTG, dashboards, alerting, logs, operational health signals
Collaboration	Independent project ownership, asynchronous written communication, architecture evaluation, documentation, team enablement

Work Experience

Énergir — Senior DevOps Engineer

Montréal, QC | September 2021 – Present

- Independently reverse-engineered nine application ecosystems—including Python, Java, Node.js, Spark, Oracle, Salesforce, MuleSoft, CloudFoundry, and Databricks—and standardized CI/CD for 12 teams and 70+ applications.
- Rebuilt Salesforce delivery with its official CLI and designed a validation service flow that deployed each candidate merge to an isolated sandbox and ran the full test suite before promotion.
- Separated application-code failures from platform failures in pipeline reporting, creating clearer system boundaries and making delivery failures self-diagnosing for product teams.
- Led Azure modernization for several mission-critical services and standardized Terraform and Ansible provisioning across development, staging, and production.
- Drove root-cause analysis on critical incidents and delivered monitoring, dashboards, and automated remediation that improved reliability and reduced recovery time.

Key technologies: Python, Java, Node.js, Microsoft Azure, Azure DevOps, Terraform, Ansible, Docker, ELK, Databricks, Salesforce

Desjardins — Senior DevOps Engineer (Contract)

Montréal, QC (Remote) | October 2025 – March 2026 — *concurrent engagement alongside the Énergir role*

- Designed and built a Python/FastAPI service that tested the configuration and health of approximately 250 remote Artifactory repositories every four hours and published results to Dynatrace.

- Turned a narrow configuration-verification request into a configurable, mostly tool-agnostic validation system that became the regression safety net for an enterprise platform upgrade.
- Designed validation logic and configurable test data spanning Docker, Helm, Node.js, Maven, Python, NuGet, and Debian repositories, keeping platform-specific concerns isolated from reusable test behaviour.
- Automated on-demand environments with Kubernetes and ArgoCD, giving administrators safe and repeatable systems for testing upgrades and configuration changes.
- Built Splunk alerts for out-of-pipeline changes and implemented lifecycle policies, improving auditability, operational safety, and platform resilience.

Key technologies: Python, FastAPI, REST APIs, Docker, Kubernetes, ArgoCD, GitHub Actions, JFrog Artifactory, Dynatrace, Splunk, Helm

Tink — DevOps Engineer

Montréal, QC | January 2018 – August 2021

- Centralized Jenkins Pipeline as Code for 80+ applications across 20+ clients, standardizing delivery across different application stacks and adding release visibility through Jira integration.
- Conceived and built a software inventory platform for 600+ servers, exposing application versions, CVE exposure, end-of-life software, and SSL/TLS risk directly to product owners.
- Built the compute, caching, database, storage, DNS, and networking foundations of an AWS migration target using EC2, S3, Route 53, VPC, self-managed caching, and a serverless relational database.
- Automated operating-system patching and kernel-update reboots across 600+ servers, reducing ongoing work to a monthly verification check with near-zero downtime.
- Built a self-healing response for an unstable Elasticsearch cluster so monitoring automatically remediated a known production failure without paging an engineer.

Key technologies: AWS, EC2, S3, Route 53, VPC, Jenkins, Bash, Linux, ELK Stack, Puppet, Spacewalk, PRTG, Jira

SOVO Technologies — Lead Technician to Software Engineer

Montréal, QC | November 2008 – December 2017

- Designed and wrote production software for approximately seven years, including the scheduling system used for the production department's complete planning cycle.
- Built a voice-recognition evaluation system and multilingual dictionary infrastructure supporting internationalized product locales.
- Owned company infrastructure, delivered a zero-downtime server relocation, and introduced redundancy that eliminated core single points of failure while managing and training technicians.

Key technologies: C++, Java, SQL, NoSQL, Linux, Windows Server, VMware vSphere, Active Directory, networking

Education

Bachelor of Automated Manufacturing Engineering — École de Technologie Supérieure, Montréal, QC | 2005 – 2016

DCS in Computer Science Technology — Cégep André-Laurendeau, Montréal, QC | 2002 – 2005